CPE 381 Classwork 06: Discrete Difference, Operations on Signals

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 14, 2026 (In-Class)

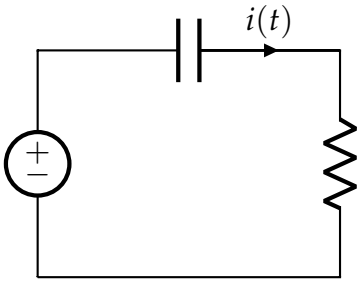
1. Taylor series expansion around operating point t_0 is given by

$$x(t_1) = x(t_0 + h) = x(t_0) + hx'(t_0) + \frac{1}{2!}h^2x''(t_0) + \cdots +$$
$$x(t_1) \approx x(t_0) + hx'(t_0)$$

where $t_1 = t_0 + h$.

Consider $x(t) = \cos(t)$, and assuming $h = 0.1$, and $t_0 = 0$, what is $\cos(t_1)$? **(5 Points)**

Answer

2. Consider a circuit 

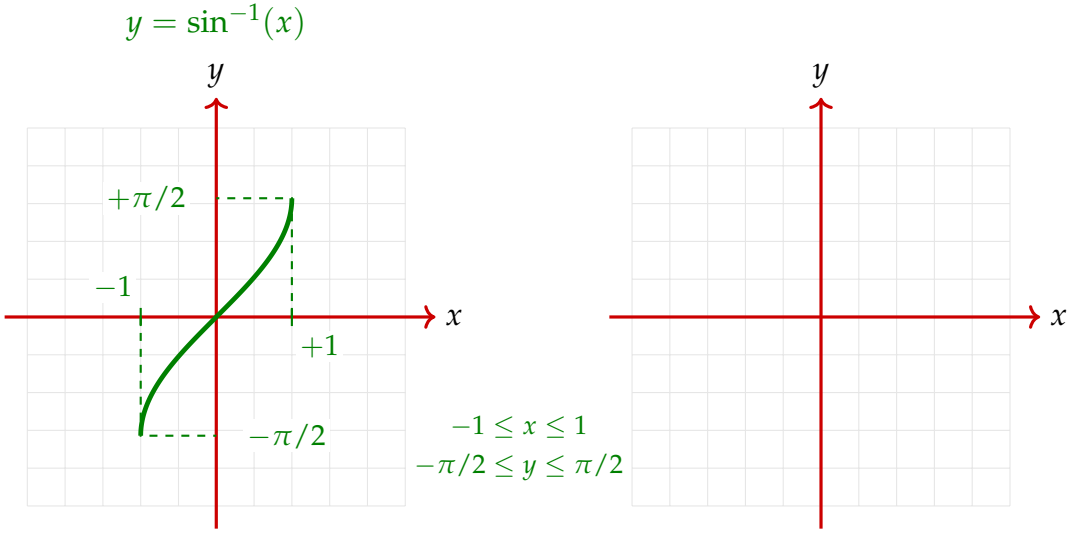
that has a constant voltage source $v_i(t)$ with $v_c(t)$ as the voltage across the capacitor. Its differential equation is given by

$$v_i(t) = v_c(t) + \frac{dv_c(t)}{dt}$$

Assume $v_c(t) = 0$ for $t \leq 0$. Further, assume Δt as the constant sampling time and write down the expression for $v_i(t)$ in the form of discrete difference equation by considering two-point backward difference as the form to compute discrete derivative. **(10 Points)**

Answer

3. $y = sin^{-1}(x)$ is given. Plot $y_1 = \sin^{-1}(x) + 1$ on the empty grid. **(2.5 Points)**



4. Consider $y = \sin(x)$. Based on this, plot $y_1 = \frac{1}{2}\sin(x)$. **(2.5 Points)**

